

MATH3881/5881: Statistical Machine Learning Theory

Week 2: Supplement: GD Convergence and the UAT

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Optional reading. This supplement fills in two proofs that the main lecture notes state but do not prove. Section 1 proves the spectral-radius claim from the Week 1 subsection *Learning Rate Analysis for Linear Regression*; Section 2 proves a constructive version of the Week 2 Universal Approximation Theorem for ReLU networks in one dimension, and sketches the general case. Nothing here is new examinable content beyond the corresponding sections of the lecture notes; students may safely skip it.

1. Why $\rho(A) < 1$ implies gradient descent converges

1.1 Setup

The Week 1 subsection *Learning Rate Analysis for Linear Regression* introduces the linear-regression iteration

$$\theta^{(t+1)} = A\theta^{(t)} + \frac{2\alpha}{n}X^\top y, \quad A := I - \frac{2\alpha}{n}X^\top X, \quad (2.1)$$

where $X \in \mathbb{R}^{n \times p}$ is the design matrix, $\alpha > 0$ the step size, and $\theta^* \in \mathbb{R}^p$ the (assumed unique) solution of the normal equations $X^\top X\theta^* = X^\top y$. The Week 1 notes state, without proof, that the iteration converges to θ^* from any starting point whenever the spectral radius $\rho(A) < 1$. We prove that implication.

1.2 Warm-up: the 1D case

To build intuition, consider the scalar version of (2.1), an affine iteration

$$x^{(t+1)} = ax^{(t)} + c$$

with $a \in \mathbb{R}$ and $a \neq 1$. The unique fixed point is $x^* = c/(1 - a)$. Subtracting the fixed-point equation $x^* = ax^* + c$ from the iteration gives $x^{(t+1)} - x^* = a(x^{(t)} - x^*)$, and iterating,

$$x^{(t)} - x^* = a^t(x^{(0)} - x^*).$$

If $|a| < 1$, then $a^t \rightarrow 0$ and therefore $x^{(t)} \rightarrow x^*$.

The matrix proof in Section 1.3 is the same argument applied component-wise in the eigenbasis of A , with the scalar a replaced by each eigenvalue μ_k of A in turn.

1.3 The matrix proof

Let $e^{(t)} := \theta^{(t)} - \theta^*$.

Step 1: error dynamics. Subtracting the fixed-point equation $\theta^* = A\theta^* + (2\alpha/n)X^\top y$ from (2.1) gives $e^{(t+1)} = Ae^{(t)}$. Iterating,

$$e^{(t)} = A^t e^{(0)}.$$

This is the matrix analog of $x^{(t)} - x^* = a^t(x^{(0)} - x^*)$ from the 1D case.

Step 2: diagonalise A . Since $X^\top X$ is symmetric, so is A . The spectral theorem gives an orthonormal basis $u_1, \dots, u_p$ of $\mathbb{R}^p$ such that $Au_k = \mu_k u_k$, with eigenvalues $\mu_k = 1 - (2\alpha/n)\lambda_k$ (where λ_k are the eigenvalues of $X^\top X$).

Step 3: decompose the initial error in this basis. Write $e^{(0)} = \sum_{k=1}^p c_k u_k$, where $c_k = e^{(0)} \cdot u_k$. Applying A^t component-wise,

$$e^{(t)} = A^t e^{(0)} = \sum_{k=1}^p c_k \mu_k^t u_k.$$

Each coordinate $c_k \mu_k^t$ is exactly the 1D iteration from Section 1.2, with $a = \mu_k$.

Step 4: each component decays. By definition of the spectral radius, $|\mu_k| \leq \rho(A) < 1$ for every k , so $\mu_k^t \rightarrow 0$ as $t \rightarrow \infty$ for every k . Hence, using the orthonormality of the u_k ,

$$\|e^{(t)}\|^2 = \sum_{k=1}^p c_k^2 \mu_k^{2t} \rightarrow 0.$$

Therefore $\theta^{(t)} = \theta^* + e^{(t)} \rightarrow \theta^*$.

2. Why ReLU networks are universal: a constructive proof in one dimension

2.1 Setup

Recall the Week 2 Universal Approximation Theorem: for any compact $\mathcal{K} \subseteq \mathbb{R}^p$, any continuous $f^* : \mathcal{K} \rightarrow \mathbb{R}^q$, any non-polynomial activation σ , and any $\varepsilon > 0$, there exist parameters such that a two-layer network f_θ approximates f^* uniformly on $\mathcal{K}$ to within ε . The original proofs (Cybenko 1989 for sigmoid via the Hahn–Banach theorem; Hornik 1991 for any non-polynomial σ via density arguments) use machinery beyond the scope of this course.

For $p = q = 1$ and $\sigma = \sigma_{\text{ReLU}}$, however, there is a direct constructive proof: build a piecewise-linear interpolant of f^* , show it can be expressed as a small ReLU network, and bound the error using uniform continuity. We give this proof in full, then sketch how the same idea extends to higher dimensions.

Theorem 2.1 (UAT for ReLU networks, 1D)

Let $f^* : [0, 1] \rightarrow \mathbb{R}$ be continuous and $\varepsilon > 0$. There exists $N \in \mathbb{N}$ and parameters $w \in \mathbb{R}^{N+1}$, $b \in \mathbb{R}^{N+1}$, $a \in \mathbb{R}^{N+1}$ such that the width- $(N+1)$ two-layer ReLU network

$$f_\theta(x) = \sum_{k=0}^N a_k \sigma_{\text{ReLU}}(w_k x + b_k)$$

satisfies $|f_\theta(x) - f^*(x)| < \varepsilon$ for every $x \in [0, 1]$.

2.2 Step 1: setup (knots and slopes)

Choose a positive integer N and equally-spaced knots

$$x_k = \frac{k}{N}, \quad k = 0, 1, \dots, N.$$

Let $y_k = f^*(x_k)$ for each k , and define the slopes

$$m_k = \frac{y_k - y_{k-1}}{x_k - x_{k-1}} = N(y_k - y_{k-1}), \quad k = 1, \dots, N.$$

We aim to construct a width- $(N+1)$ two-layer ReLU network that agrees with f^* at each knot x_k and is affine on each subinterval $[x_{k-1}, x_k]$.

2.3 Step 2: define the network and check interpolation

Define the width- $(N+1)$ ReLU network

$$f_\theta(x) = y_0 \sigma_{\text{ReLU}}(0 \cdot x + 1) + m_1 \sigma_{\text{ReLU}}(x) + \sum_{k=2}^N (m_k - m_{k-1}) \sigma_{\text{ReLU}}(x - x_{k-1}), \quad (2.2)$$

with hidden-layer parameters and output weights

$$(w_0, b_0, a_0) = (0, 1, y_0), \quad w_k = 1, \quad b_k = -x_{k-1} \quad (k = 1, \dots, N), \quad a_1 = m_1, \quad a_k = m_k - m_{k-1} \quad (k \geq 2).$$

The first term, $y_0 \sigma_{\text{ReLU}}(1) = y_0$, contributes the constant y_0 at every x , absorbing what would otherwise be an output bias into a hidden unit with input weight zero. So the network matches the Week 2 form $W^{[2]} \sigma(W^{[1]} x + b^{[1]})$ exactly.

We now verify directly that f_θ is the piecewise-linear interpolant of f^* at the knots $\{x_0, \dots, x_N\}$, that is, $f_\theta(x_k) = y_k$ for every k and f_θ is affine on each $[x_{j-1}, x_j]$.

Affineness and slope on $[x_{j-1}, x_j]$. Fix $j \in \{1, \dots, N\}$ and $x \in (x_{j-1}, x_j)$. The active ReLU units in (2.2) are those with positive argument: the constant unit (always active); the $m_1 \sigma_{\text{ReLU}}(x)$ unit

(since $x > 0$); and the units $k = 2, \dots, j$ (since $x > x_{k-1}$ for these k). The units with $k > j$ are zero because $x < x_{k-1}$. Hence

$$f_\theta(x) = y_0 + m_1 x + \sum_{k=2}^j (m_k - m_{k-1})(x - x_{k-1}),$$

which is affine in x . Its slope on (x_{j-1}, x_j) is

$$m_1 + \sum_{k=2}^j (m_k - m_{k-1}) = m_j,$$

where the sum telescopes.

Values at the knots. At $x = 0$, only the constant unit is active: $f_\theta(0) = y_0$. For $j \geq 1$, f_θ is affine on $[x_{j-1}, x_j]$ with slope m_j , so

$$f_\theta(x_j) = f_\theta(x_{j-1}) + m_j(x_j - x_{j-1}) = f_\theta(x_{j-1}) + (y_j - y_{j-1}).$$

Induction on j gives $f_\theta(x_j) = y_j$ for every j .

So f_θ matches f^* at every knot and is affine between consecutive knots.

2.4 Step 3: bound the approximation error

Since f^* is continuous on the compact set $[0, 1]$, it is uniformly continuous: for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|x - y| < \delta$ implies $|f^*(x) - f^*(y)| < \varepsilon$.

Step 3a. Fix $x \in [0, 1]$ and choose j with $x \in [x_{j-1}, x_j]$. By Step 2, f_θ is affine on $[x_{j-1}, x_j]$ with $f_\theta(x_{j-1}) = y_{j-1} = f^*(x_{j-1})$ and $f_\theta(x_j) = y_j = f^*(x_j)$. Hence $f_\theta(x)$ is a convex combination of $f^*(x_{j-1})$ and $f^*(x_j)$: $f_\theta(x) = (1 - t)f^*(x_{j-1}) + tf^*(x_j)$ for some $t \in [0, 1]$.

Step 3b. By the triangle inequality and the convex combination,

$$|f_\theta(x) - f^*(x)| \leq (1 - t)|f^*(x_{j-1}) - f^*(x)| + t|f^*(x_j) - f^*(x)| \leq \max_{y \in [x_{j-1}, x_j]} |f^*(y) - f^*(x)|.$$

Both x and the maximising y lie in $[x_{j-1}, x_j]$, so $|x - y| \leq 1/N$. Therefore

$$|f_\theta(x) - f^*(x)| \leq \omega_{f^*}(1/N), \quad \omega_{f^*}(\delta) := \sup_{|x-y| \leq \delta} |f^*(x) - f^*(y)|,$$

where ω_{f^*} is the modulus of continuity of f^* .

Step 3c. By uniform continuity, $\omega_{f^*}(\delta) \rightarrow 0$ as $\delta \rightarrow 0^+$. Choose N large enough that $\omega_{f^*}(1/N) < \varepsilon$. Then $|f_\theta(x) - f^*(x)| < \varepsilon$ for every $x \in [0, 1]$, completing the proof of Theorem .

2.5 Sketch of the general case

For general dimensions $p, q \geq 1$ and a general non-polynomial activation σ , the same strategy works in principle but requires more bookkeeping.

1. *Reduce to scalar outputs.* It is enough to prove the theorem for $q = 1$, because each output coordinate of $f^* : \mathcal{K} \rightarrow \mathbb{R}^q$ can be approximated independently and the resulting networks can be combined by stacking output weights.
2. *Reduce to a hypercube.* Compactness lets us enclose $\mathcal{K}$ in a hypercube $[a, b]^p$; if a network approximates f^* on the hypercube it approximates it on $\mathcal{K}$.
3. *Triangulate and interpolate.* The hypercube can be subdivided into a regular triangulation by simplices of diameter at most δ . The linear interpolant $\hat{f}$ of f^* at the simplex vertices is a continuous piecewise-linear function whose error is bounded by the multidimensional modulus of continuity of f^* , $\omega_{f^*}(\delta)$, which goes to 0 as $\delta \rightarrow 0$ by uniform continuity.
4. *Express the interpolant as a ReLU network.* Any continuous piecewise-linear function $\hat{f} : \mathbb{R}^p \rightarrow \mathbb{R}$ can be written as a difference of two convex piecewise-linear functions, and each convex piecewise-linear function is the maximum of finitely many affine functions. The identity $\max(a, b) = a + \sigma_{\text{ReLU}}(b - a)$ then lets us express each such maximum, and hence $\hat{f}$, as a (two- or three-layer) ReLU network. Hanin (2019)¹ and Yarotsky (2017)² make this explicit and give quantitative bounds on the required width.
5. *Non-ReLU activations.* For a general non-polynomial σ , the result follows from Hornik (1991): the relevant fact is that any non-polynomial activation is dense (after sums of shifts and scales) in the space of continuous functions on a compact interval; chaining with the piecewise-linear reduction recovers the general case.

Remark 2.0.1

None of the steps above gives any indication of how large N_1 needs to be in practice. The 1D bound $\omega_{f^*}(1/N) < \varepsilon$ tells us that for a Lipschitz f^* with constant L we may take $N \geq L/\varepsilon$, but for less regular f^* (or higher p), the required width can blow up. The UAT is purely an existence result; concrete training is a separate matter handled by gradient descent (Weeks 3–4).

¹B. Hanin. *Universal function approximation by deep neural nets with bounded width and ReLU activations*. Mathematics, 7(10):992, 2019.

²D. Yarotsky. *Error bounds for approximations with deep ReLU networks*. Neural Networks, 94:103–114, 2017.